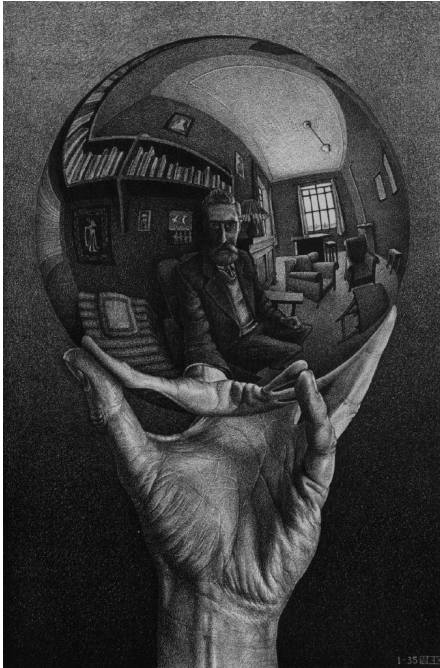


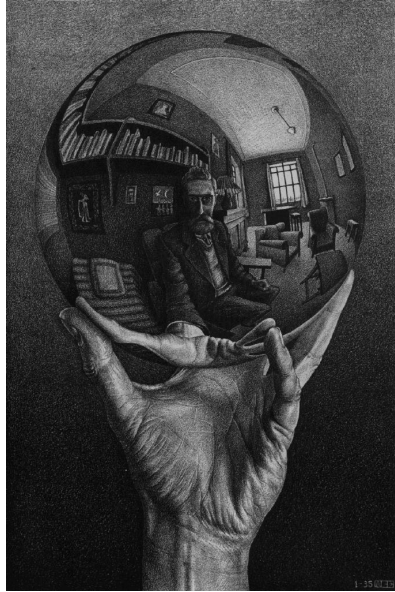
Catadioptric Image Formation & Event Cameras



S.K. Nayar and S. Baker

S.K. Nayar and A. Karmarkar

- What are Catadioptric Systems ?
- How to implement the idea with conventional cameras?
- What is the resolution of the resulting systems?

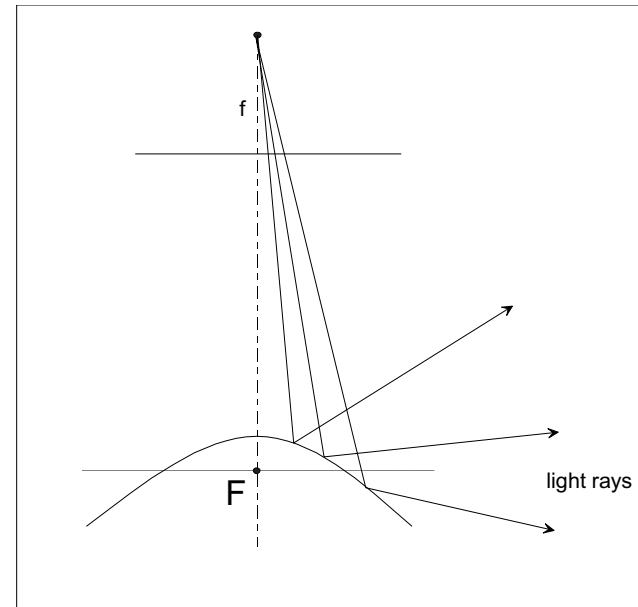
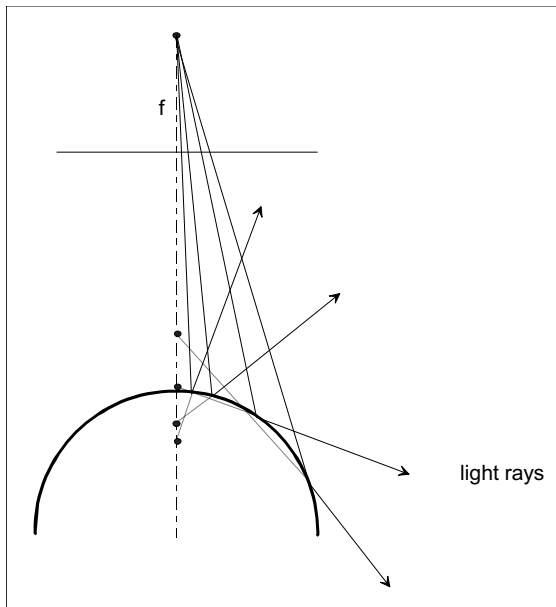


Catadioptric Cameras

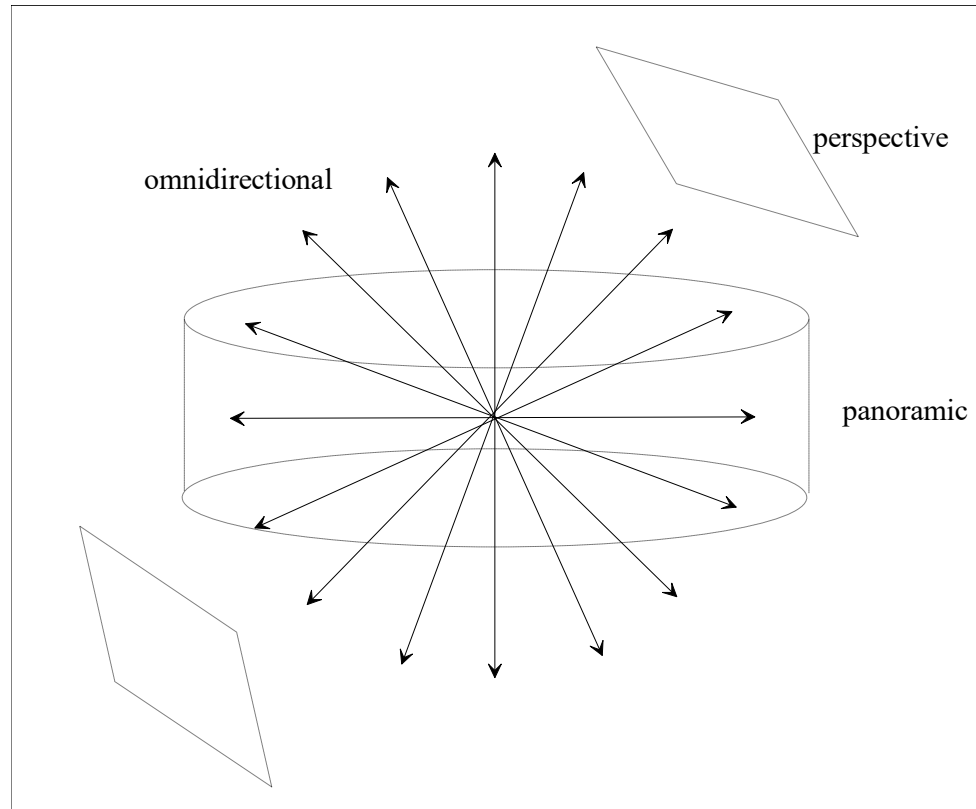
Dioptrics – science of refracting elements (lenses)

Catoptrics – science of reflecting mirrors

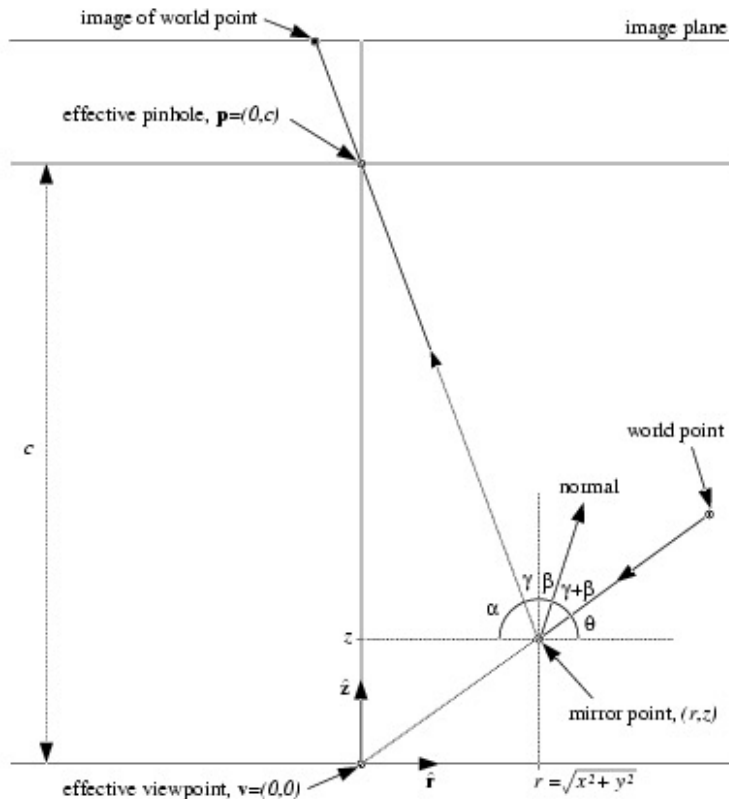
Catadioptrics – combination of refracting
and reflecting elements



Single Viewpoint Projection



Geometry used to derive the reflecting surface



Surface needs to satisfy the following equation for single viewpoint

$$\frac{-2\frac{dz}{dr}}{1 - \left(\frac{dz}{dr}\right)^2} = \frac{(c - 2z)r}{r^2 + cz - z^2}$$

Resulting in following quadric differential equation

$$r(c - 2z) \left(\frac{dz}{dr}\right)^2 - 2(r^2 + cz - z^2) \frac{dz}{dr} + r(2z - c) = 0.$$

General solution of the Constrain Equation

$$\frac{dz}{dr} = \frac{(z^2 - r^2 - cz) \pm \sqrt{r^2 c^2 + (z^2 + r^2 - cz)^2}}{r(2z - c)}.$$

The next step is to substitute $y = z - \frac{c}{2}$ and set $b = \frac{c}{2}$ which yields:

$$\frac{dy}{dr} = \frac{(y^2 - r^2 - b^2) \pm \sqrt{4r^2 b^2 + (y^2 + r^2 - b^2)^2}}{2ry}.$$

Then, we substitute $2rx = y^2 + r^2 - b^2$, which when differentiated gives:

$$2y \frac{dy}{dr} = 2x + 2r \frac{dx}{dr} - 2r$$

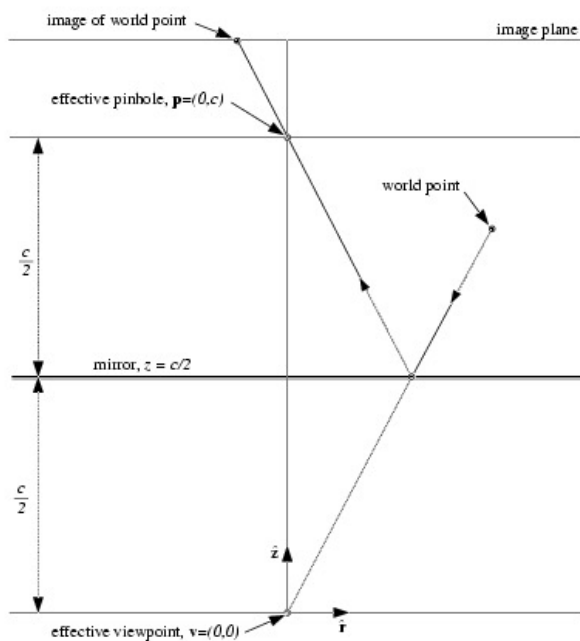
$$\left(z - \frac{c}{2}\right)^2 - r^2 \left(\frac{k}{2} - 1\right) = \frac{c^2}{4} \left(\frac{k-2}{k}\right) \quad (k \geq 2)$$

$$\left(z - \frac{c}{2}\right)^2 + r^2 \left(1 + \frac{c^2}{2k}\right) = \left(\frac{2k + c^2}{4}\right) \quad (k > 0).$$

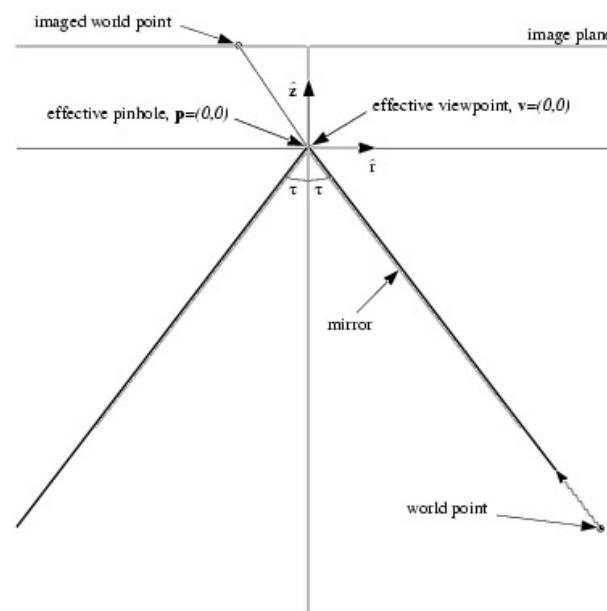
Specific mirrors

$$\left(z - \frac{c}{2}\right)^2 - r^2 \left(\frac{k}{2} - 1\right) = \frac{c^2}{4} \left(\frac{k-2}{k}\right) \quad (k \geq 2)$$

$$\left(z - \frac{c}{2}\right)^2 + r^2 \left(1 + \frac{c^2}{2k}\right) = \left(\frac{2k + c^2}{4}\right) \quad (k > 0).$$



$k=2$

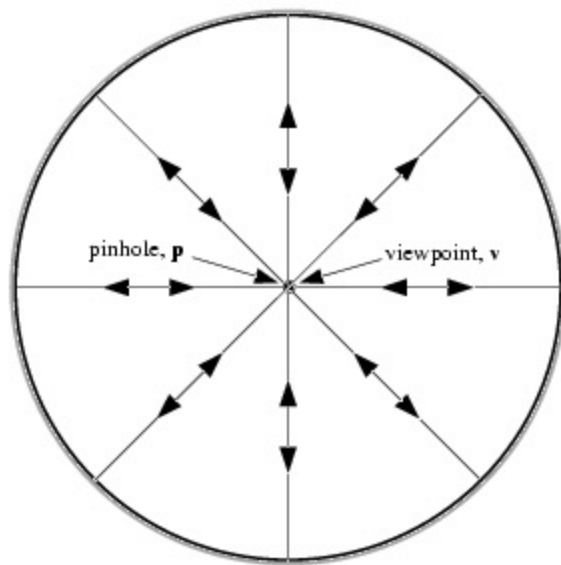


$c=0, k \geq 2$

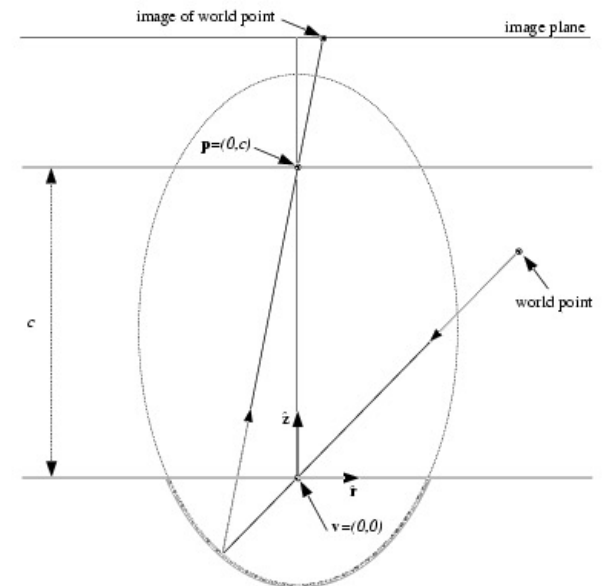
Specific mirrors (2)

$$\left(z - \frac{c}{2}\right)^2 - r^2 \left(\frac{k}{2} - 1\right) = \frac{c^2}{4} \left(\frac{k-2}{k}\right) \quad (k \geq 2)$$

$$\left(z - \frac{c}{2}\right)^2 + r^2 \left(1 + \frac{c^2}{2k}\right) = \left(\frac{2k + c^2}{4}\right) \quad (k > 0).$$

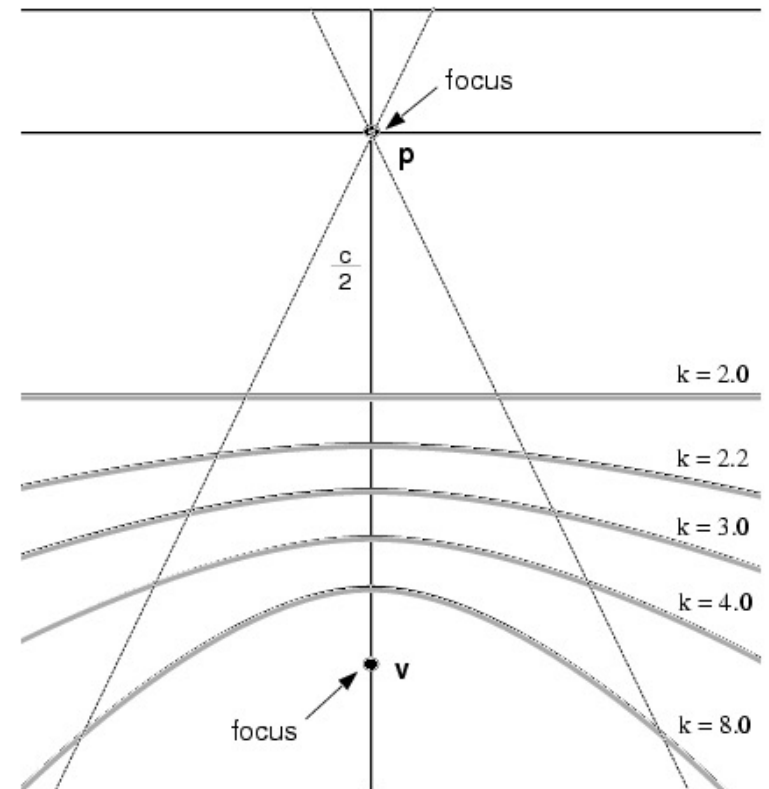
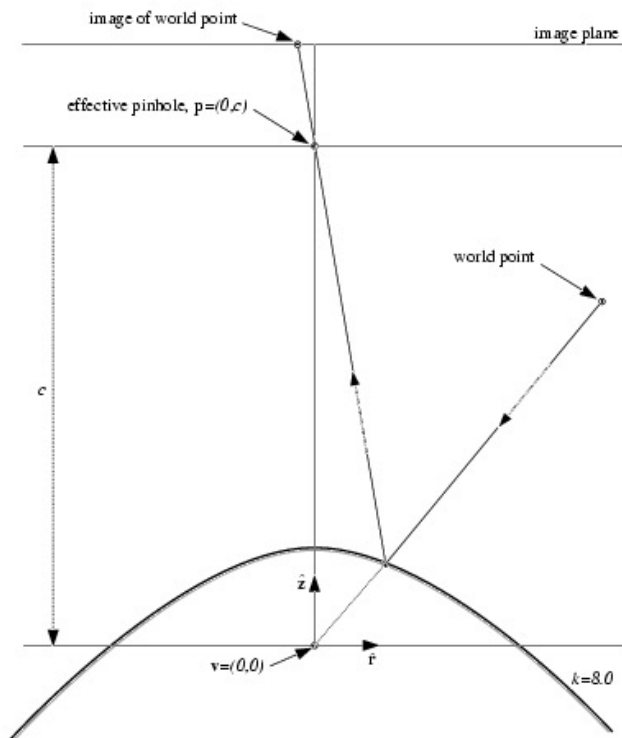


$c=0, k>0$



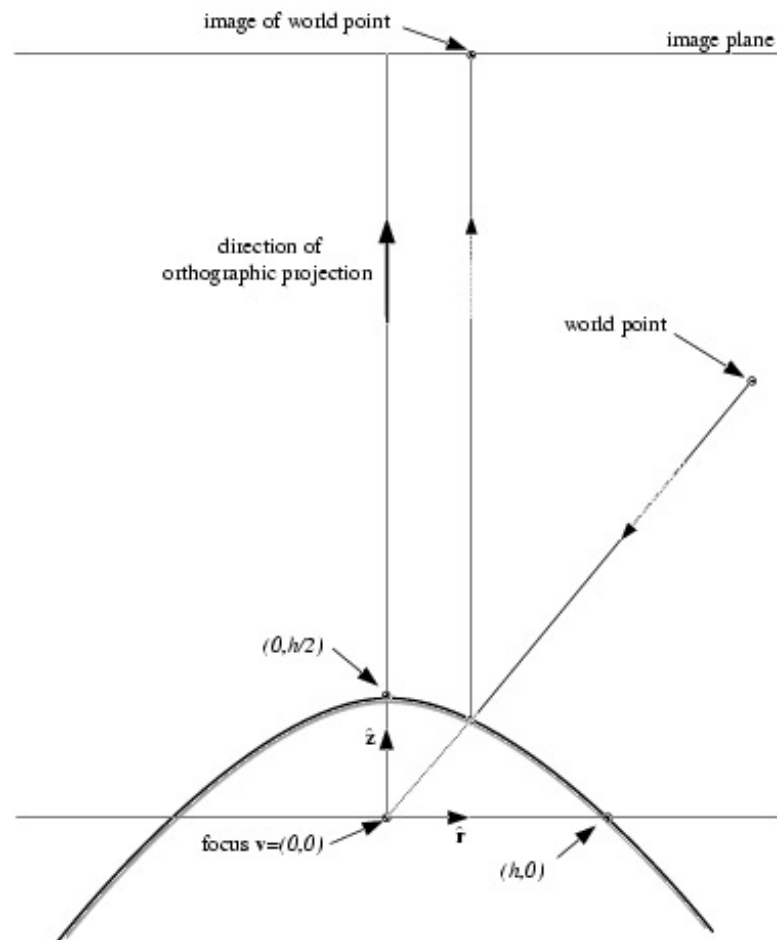
$c>0, k>0$

Specific mirrors (3)

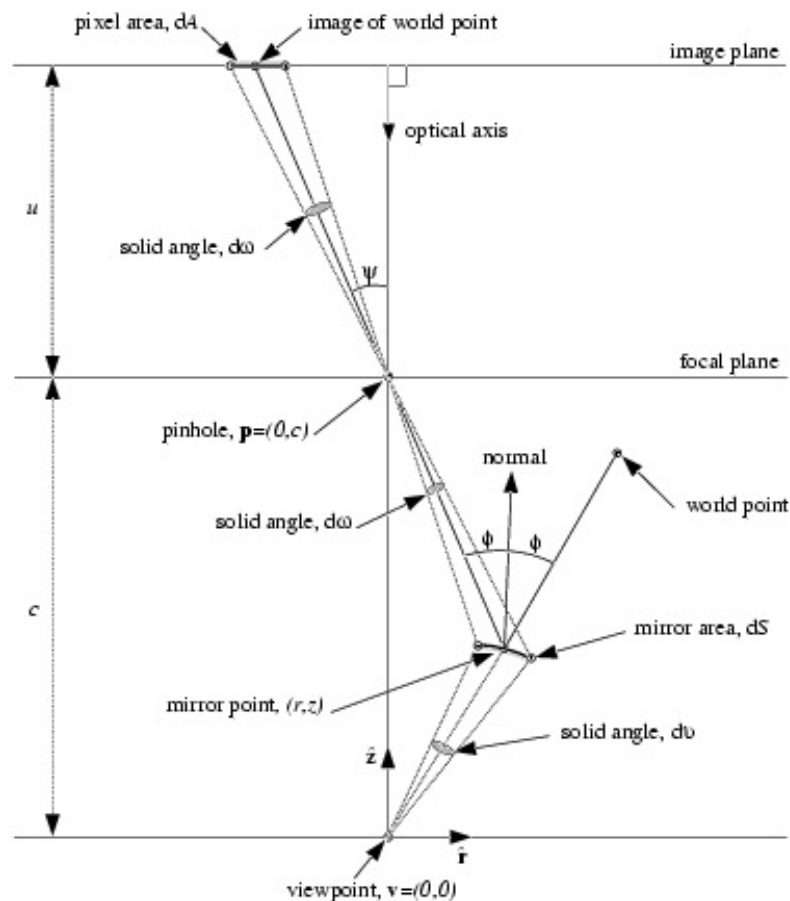


$c>0, k>2$

Orthographic projection



Resolution of the system



$$\frac{dA}{d\nu} = \frac{u^2(r^2 + z^2)}{(c - z)^2 \cdot \cos \psi} = \left[\frac{(r^2 + z^2) \cos^2 \psi}{(c - z)^2} \right] \frac{dA}{d\omega}.$$

$$\cos^2 \psi = \frac{(c - z)^2}{(c - z)^2 + r^2}$$

$$\frac{dA}{d\nu} = \left[\frac{r^2 + z^2}{(c - z)^2 + r^2} \right] \frac{dA}{d\omega}.$$

⇒ For planar mirror both resolutions are equal

Open Challenges in Computer Vision

The past 60 years of research have been devoted to frame-based cameras ...but they are not good enough!

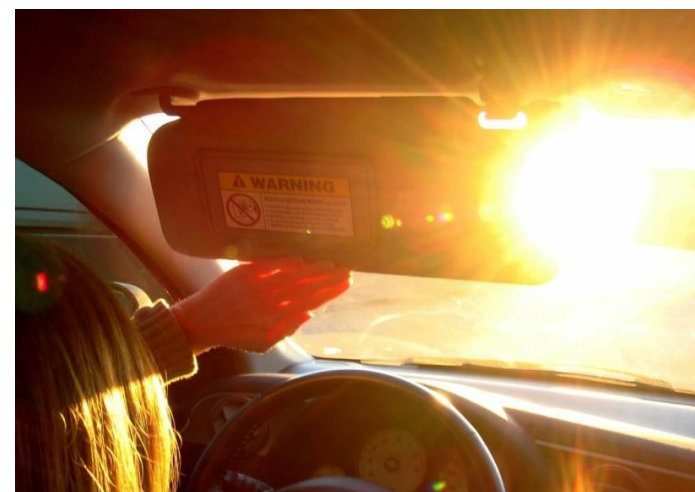
Latency



Motion blur



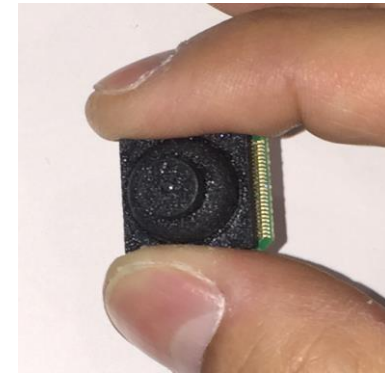
Dynamic Range



Event cameras do not suffer from these problems!

What is an event camera?

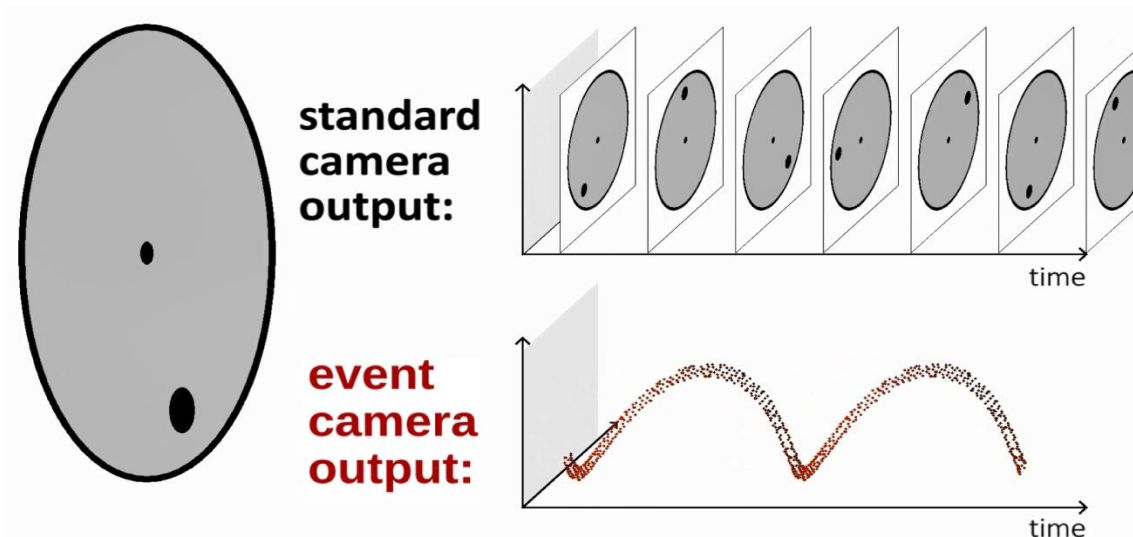
- Novel sensor that measures only **motion in the scene**
- **First commercialized in 2008** by T. Delbruck (UZHÐ) under the name of Dynamic Vision Sensor (DVS)
- **Low-latency** ($\sim 1 \mu\text{s}$)
- **No motion blur**
- **High dynamic range** (140 dB instead of 60 dB)
- **Ultra-low power** (mean: 1mW vs 1W)



Mini DVS sensor from IniVation.com

Traditional vision algorithms cannot be used because:

- **Asynchronous** pixels
- **No intensity information** (only binary intensity changes)



Video from here: <https://youtu.be/LauQ6LWTkxM?t=30>

What is an event camera?

- Novel sensor that measures only **motion in the scene**
- **First commercialized in 2008** by T. Delbruck (UZHÐ) under the name of Dynamic Vision Sensor (DVS)
- **Low-latency** ($\sim 1 \mu\text{s}$)
- **No motion blur**
- **High dynamic range** (140 dB instead of 60 dB)
- **Ultra-low power** (mean: 1mW vs 1W)

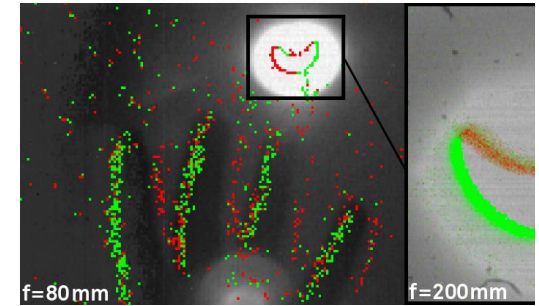
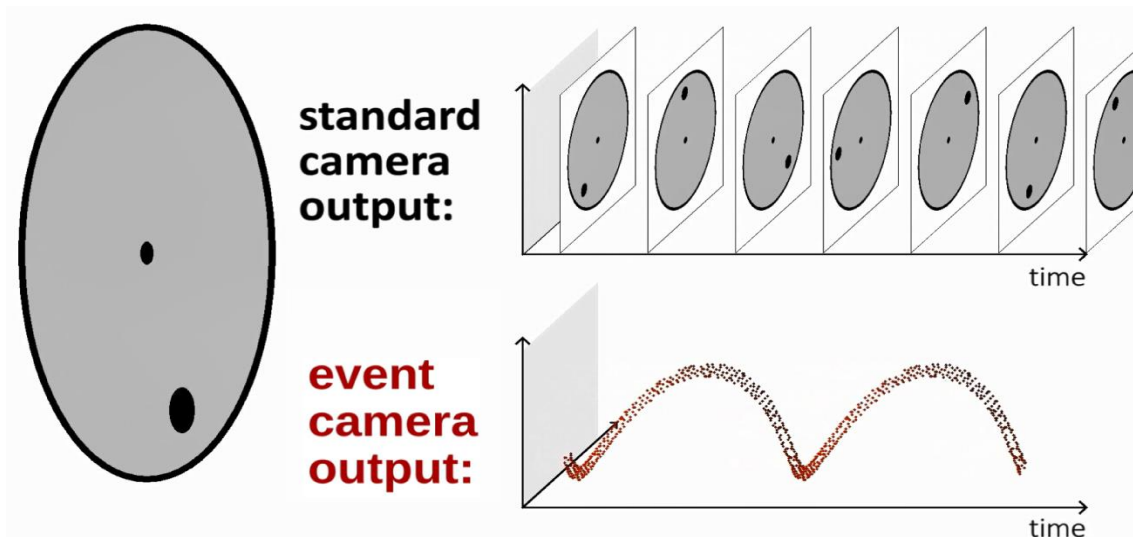


Image of the solar eclipse captured by a DVS

Traditional vision algorithms cannot be used because:

- **Asynchronous** pixels
- **No intensity information** (only binary intensity changes)



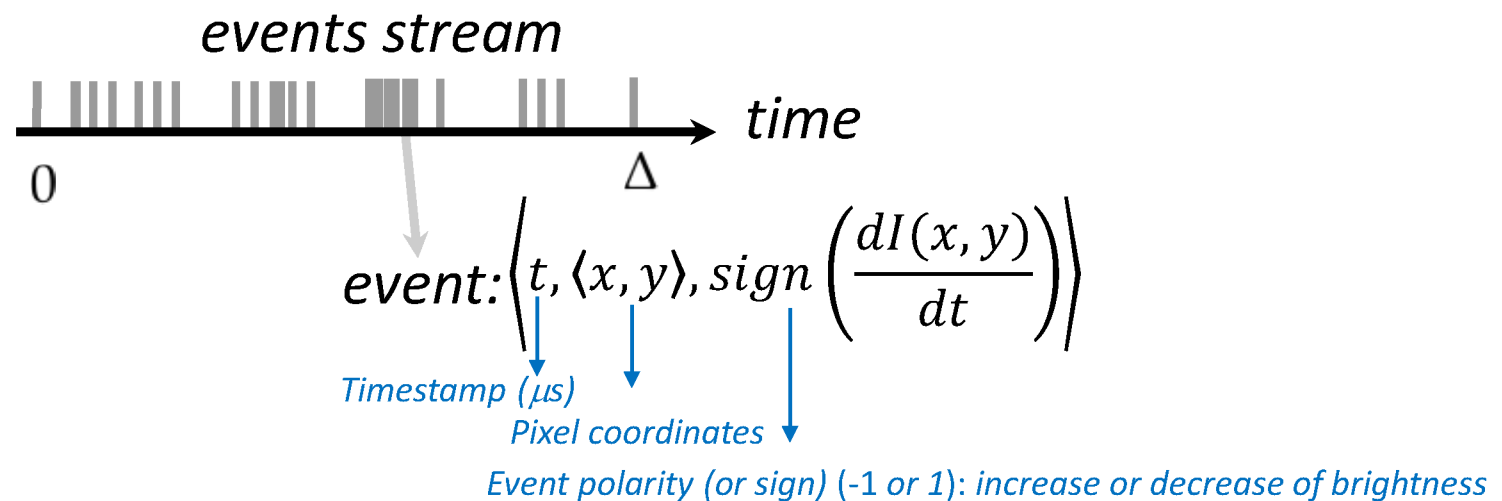
Video from here: <https://youtu.be/LauQ6LWTkxM?t=30>

Camera vs Event Camera

- A **traditional camera** outputs frames at **fixed time intervals**:



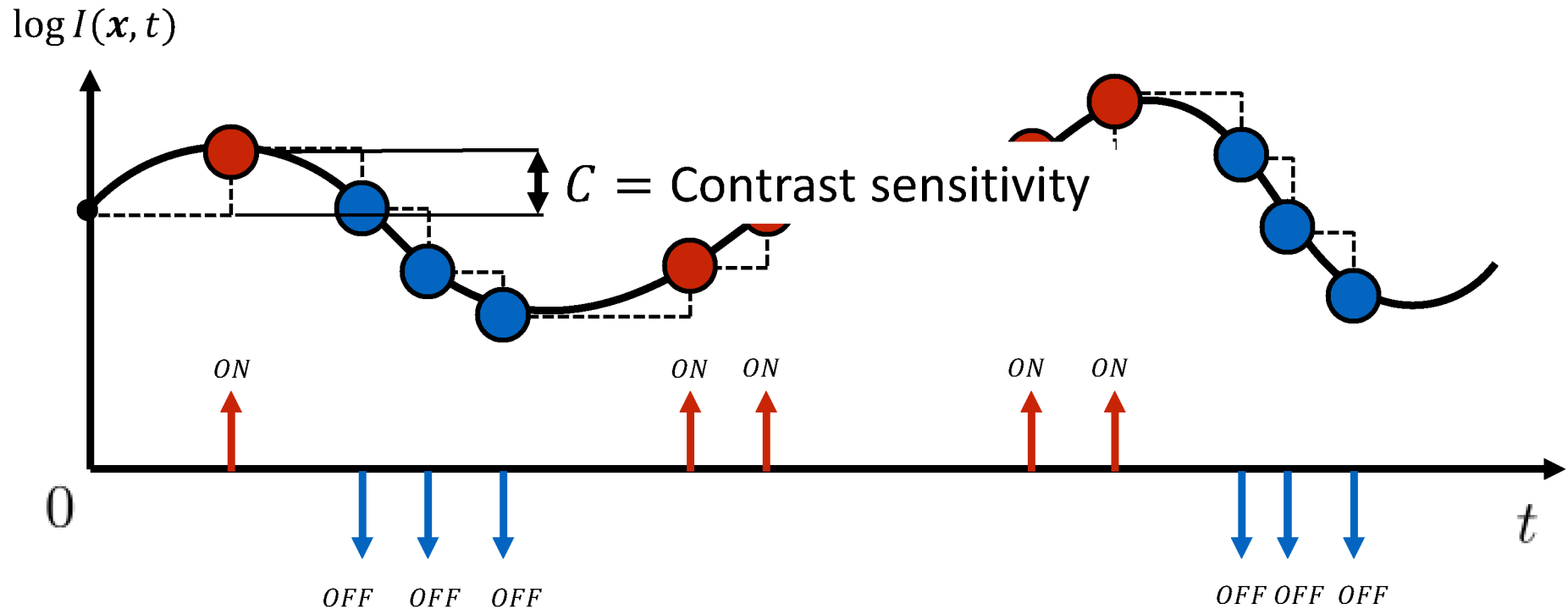
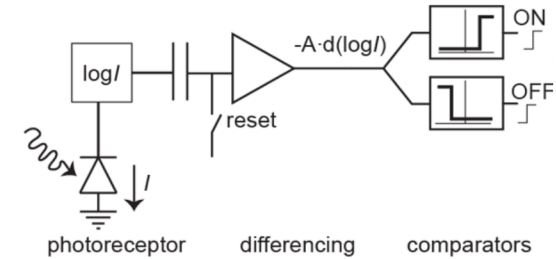
- By contrast, a **DVS** outputs **asynchronous events** at **microsecond resolution**. An event is generated each time a single pixel detects an intensity changes value



Generative Event Model

Consider the intensity at a **single pixel**...

$$\pm C = \log I(x, t) - \log I(x, t - \Delta t)$$

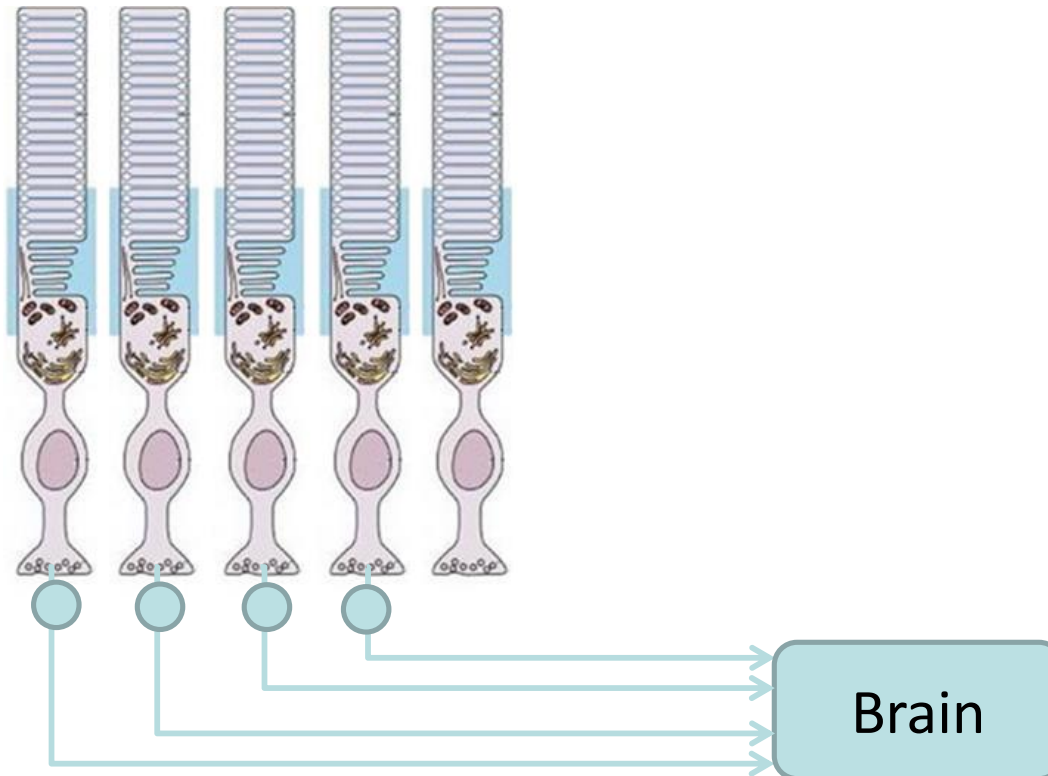
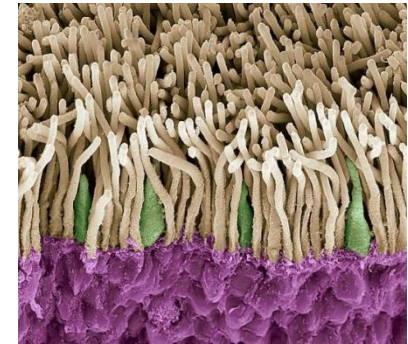
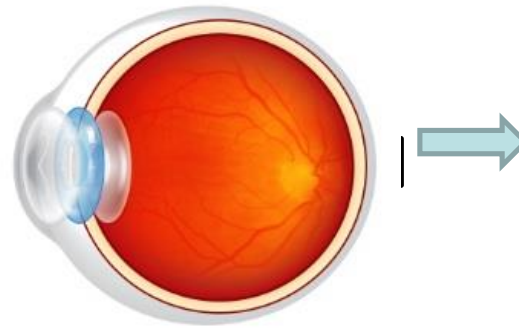


Events are triggered **asynchronously**

Event cameras are inspired by the Human Eye

Human retina:

- 130 million **photoreceptors**
- But only 2 million **axons**!



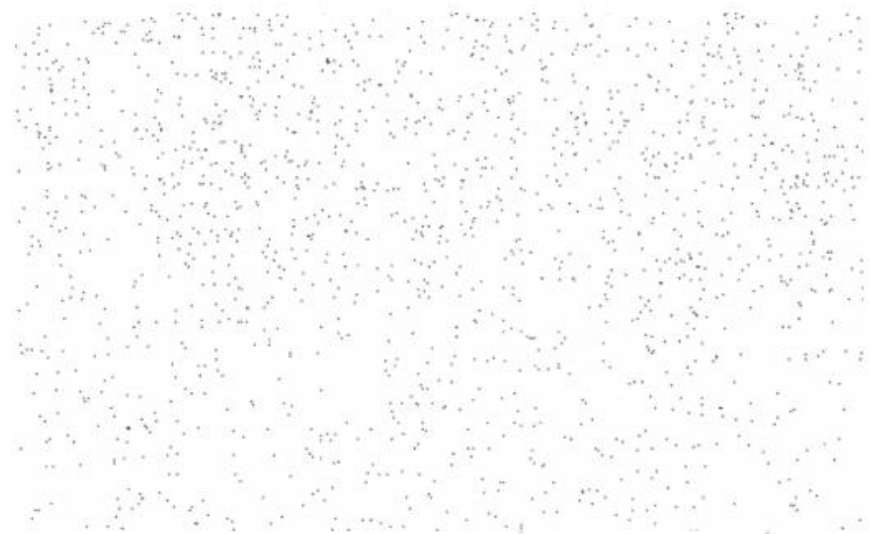
Event Camera with Static Motion

Without motion, only background noise is output

Standard Camera



Event Camera (ON, OFF events)



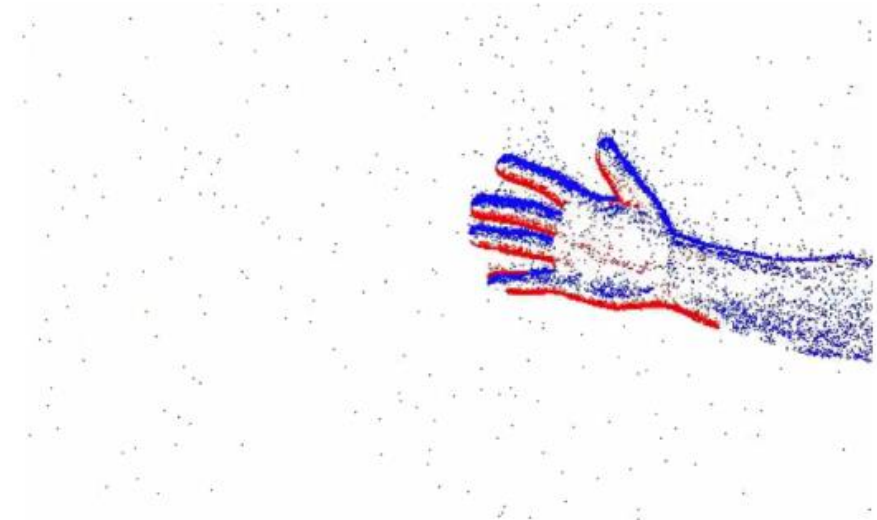
$\Delta T = 40 \text{ ms}$

Event Camera output with Motion

Standard Camera



Event Camera (ON, OFF events)



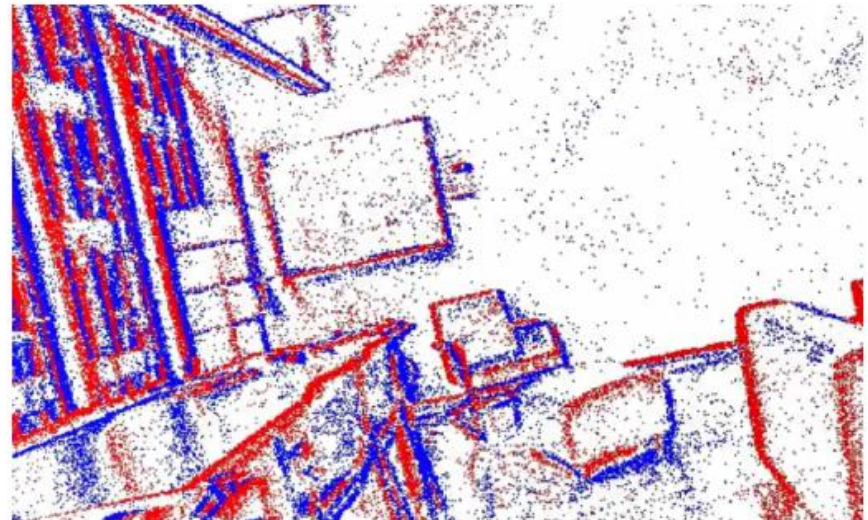
$\Delta T = 10 \text{ ms}$

Event Camera output with Motion

Standard Camera



Event Camera (ON, OFF events)

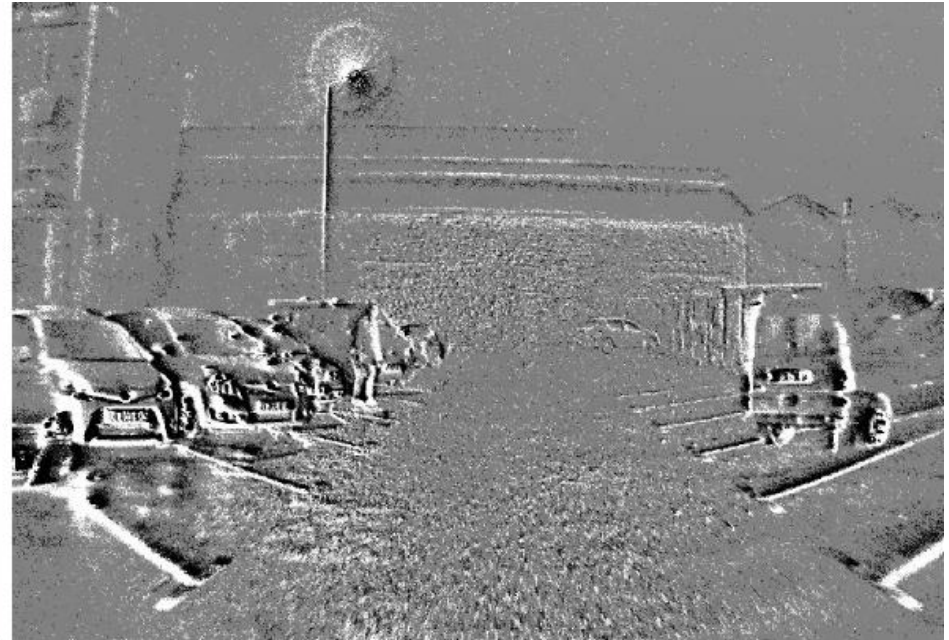


$\Delta T = 40 \text{ ms}$

Low-light Sensitivity (night drive)



GoPro Hero 6



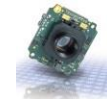
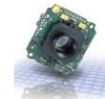
Event Camera by Prophesee

White = Positive events

Black = Negative events

Video courtesy of Prophesee: <https://www.prophesee.ai>

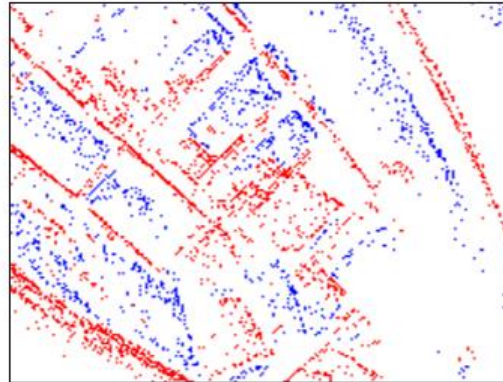
High-speed vs Event Cameras



	High speed camera	Standard camera	Event Camera
Max fps or measurement rate	Up to 1MHz	100-1,000 fps	1MHz
Resolution at max fps	64x16 pixels	>1Mpxl	>1Mpxl
Bits per pixels (event)	12 bits	8-10 per pixel	~40 bits/event {t,(x,y),p}
Weight	6.2 Kg	30 g	30 g
Active cooling	yes	No cooling	No cooling
Data rate	1.5 GB/s	32MB/s	~1MB/s on average (depends on dynamics)
Mean power consumption	150 W + external light	1 W	1 mW
Dynamic range	n.a.	60 dB	140 dB

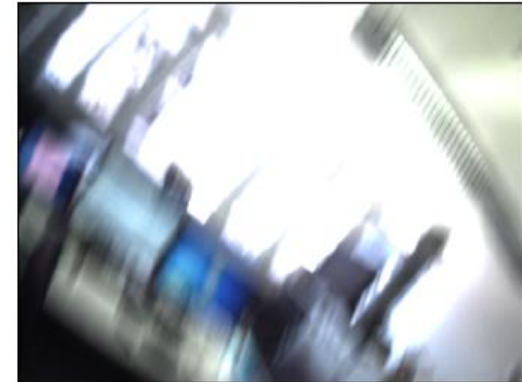
Why combining them?

< 10 years research



Event Camera

> 60 years of research!

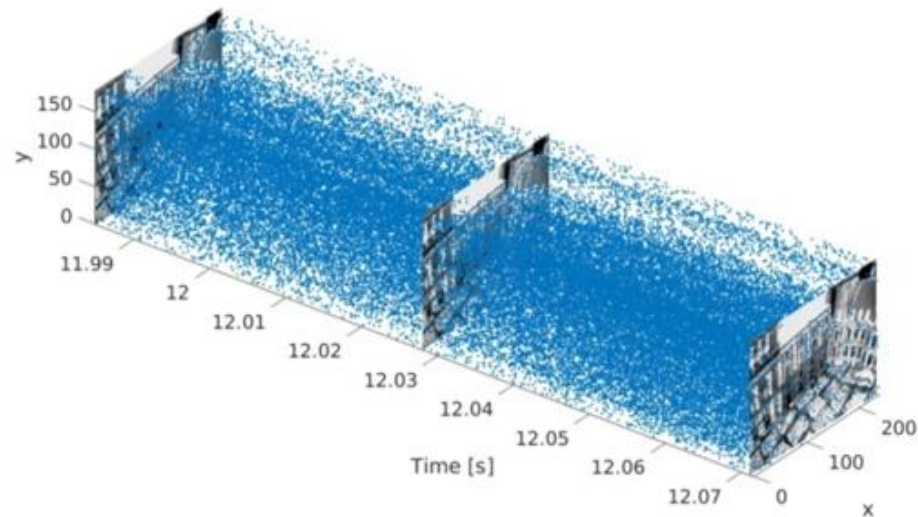


Standard Camera

Update rate	High (asynchronous): 1 MHz	Low (synchronous)
Dynamic Range	High (140 dB)	Low (60 dB)
Motion Blur	No	Yes
Static motion	No (event camera is a high pass filter)	Yes
Absolute intensity	No (reconstructable up to a constant)	Yes
Maturity	< 10 years of research	> 60 years of research!

DAVIS sensor: Events + Images + IMU

- Combines an **event** and a **standard** camera in **the same pixel array** (→ the same pixel can both trigger events and integrate light intensity).
- It also has an IMU



Standard images



Events



time

Example 1: Deblurring a blurry video

- A **blurry image** can be regarded as the **integral of a sequence of *latent images*** during the exposure time, while the **events** indicate the **changes between the latent images**.
- **Finding:** sharp image obtained by subtracting the double integral of event from input image

$$\log \left[\text{Input blur image} \right] - \iint \left[\text{Input events} \right] = \log \left[\text{Output sharp image} \right]$$

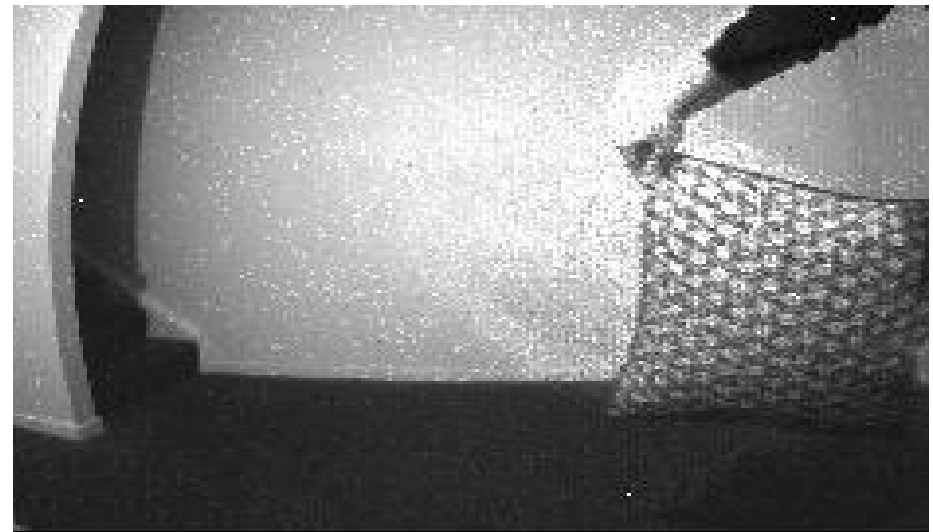
Input blur image **Input events** **Output sharp image**

Example 1: Deblurring a blurry video

- A **blurry image** can be regarded as the **integral of a sequence of *latent images*** during the exposure time, while the **events** indicate the **changes between the latent images**.
- **Finding:** sharp image obtained by subtracting the double integral of event from input image



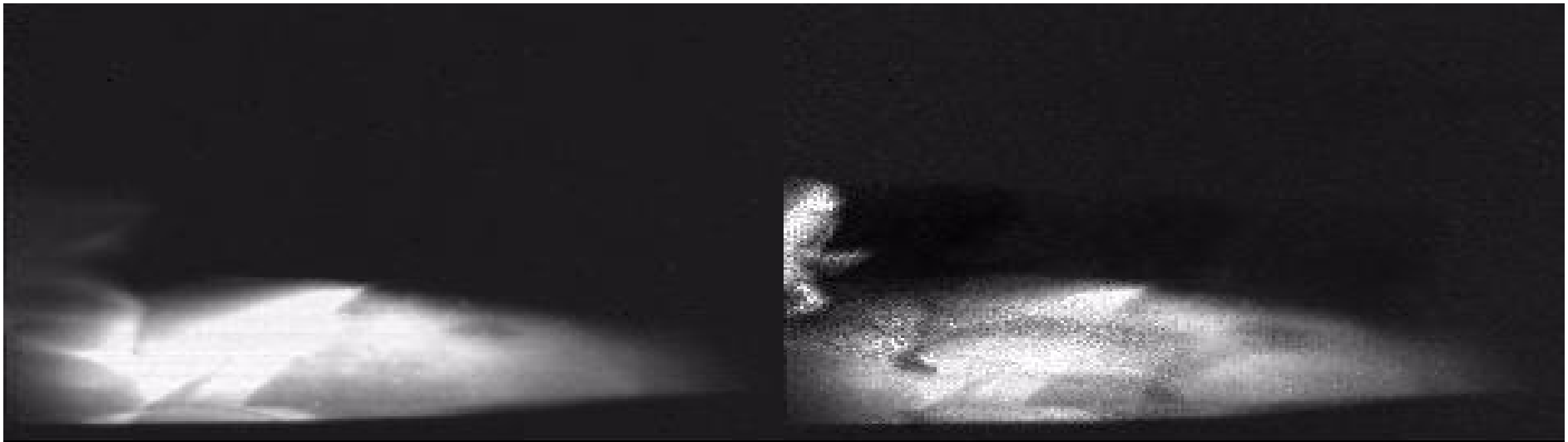
Input blur image



Output sharp video

Example 1: Deblurring a blurry video

- A **blurry image** can be regarded as the **integral of a sequence of *latent images*** during the exposure time, while the **events** indicate the **changes between the latent images**.
- **Finding:** sharp image obtained by subtracting the double integral of event from input image



Input blur image

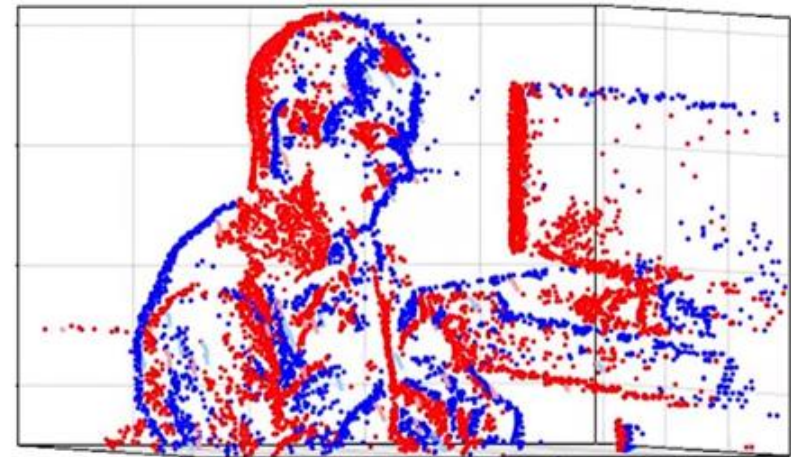
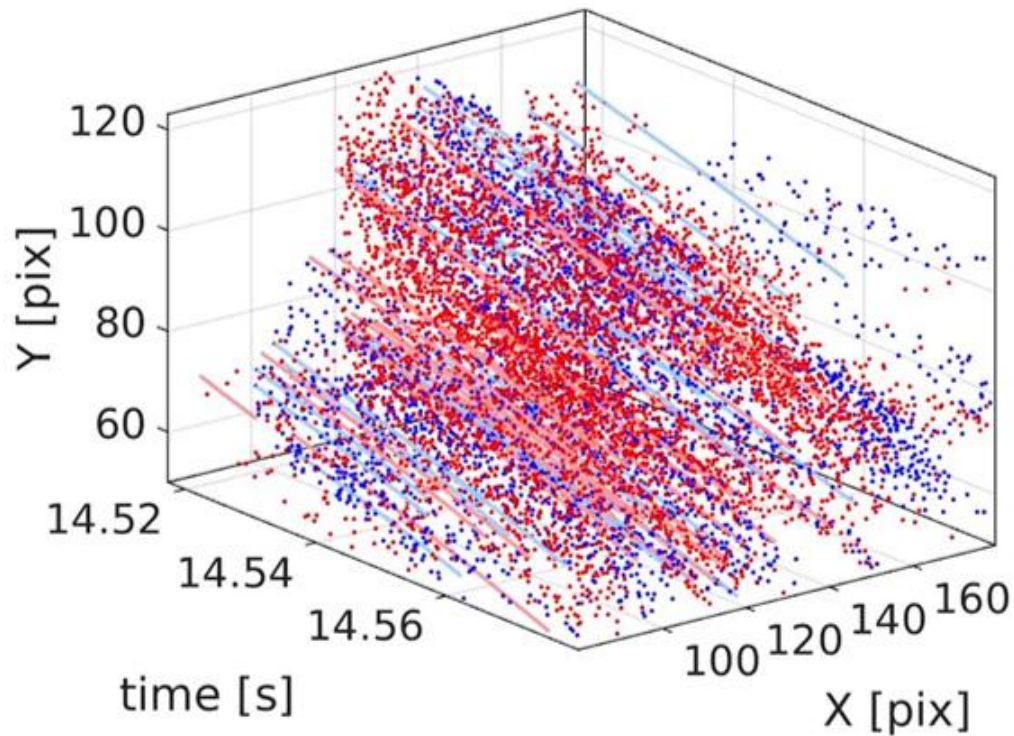
Output sharp video

Focus Maximization for:

- Motion estimation
- 3D reconstruction
- SLAM
- Optical flow estimation
- Feature tracking
- Motion segmentation
- Unsupervised learning

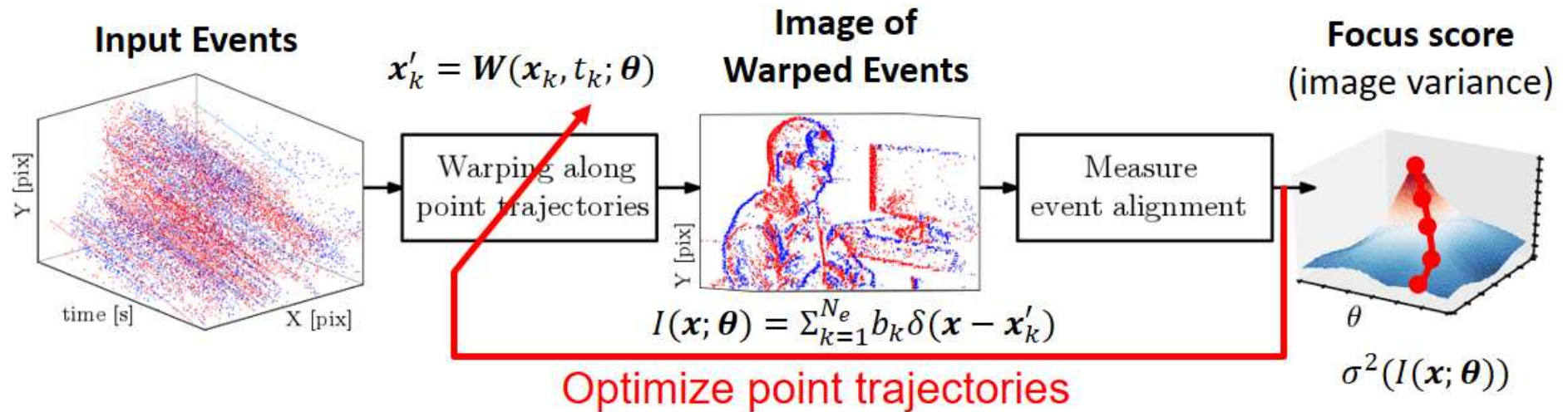
Gallego et al., A Unifying Contrast Maximization Framework for Event Cameras, CVPR18, [PDF](#), [YouTube](#)
Gallego et al., Focus Is All You Need: Loss Functions for Event-based Vision, CVPR19, [PDF](#).

Idea: Warp spatio-temporal volume of events to **maximize focus** (e.g., sharpness) of the resulting image



Gallego et al., A Unifying Contrast Maximization Framework for Event Cameras, CVPR18, [PDF](#), [YouTube](#)
Gallego et al., Focus Is All You Need: Loss Functions for Event-based Vision, CVPR19, [PDF](#).

Focus Maximization Framework

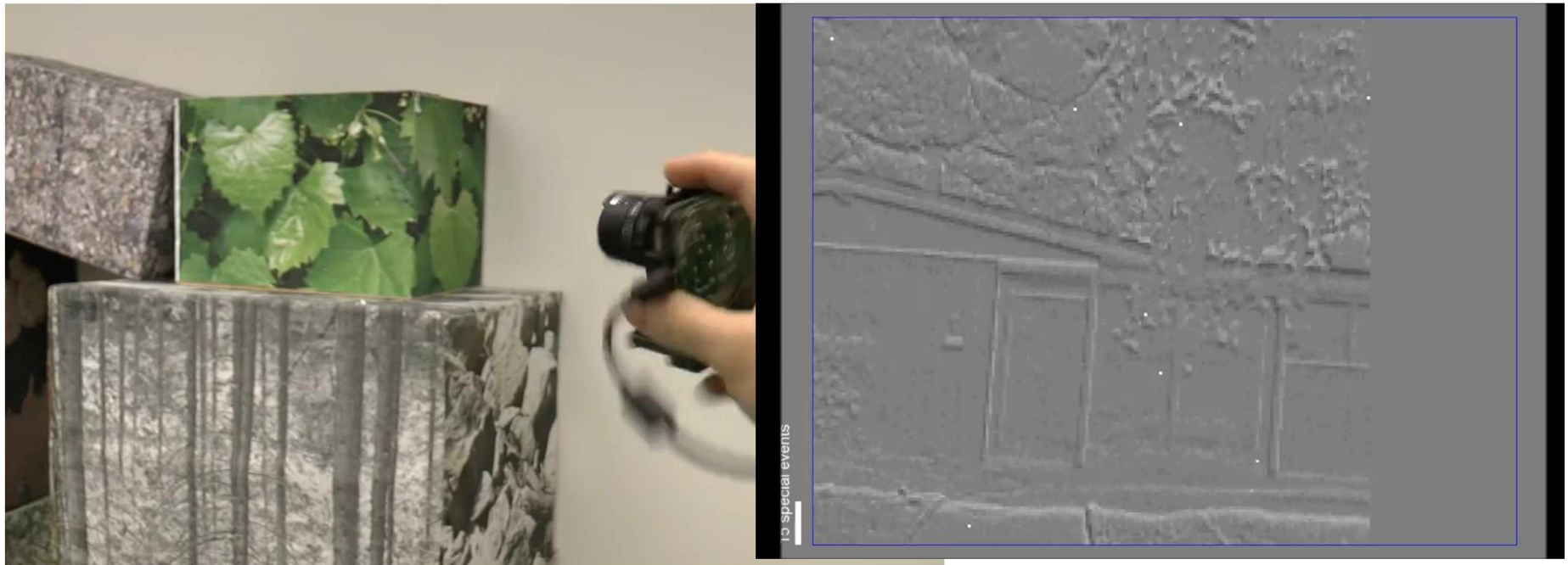
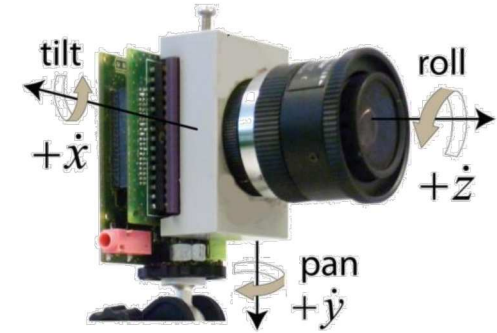


Can be implemented in a sliding-window fashion to
enable per low-latency, per-event update rate

Runs in real time on a CPU

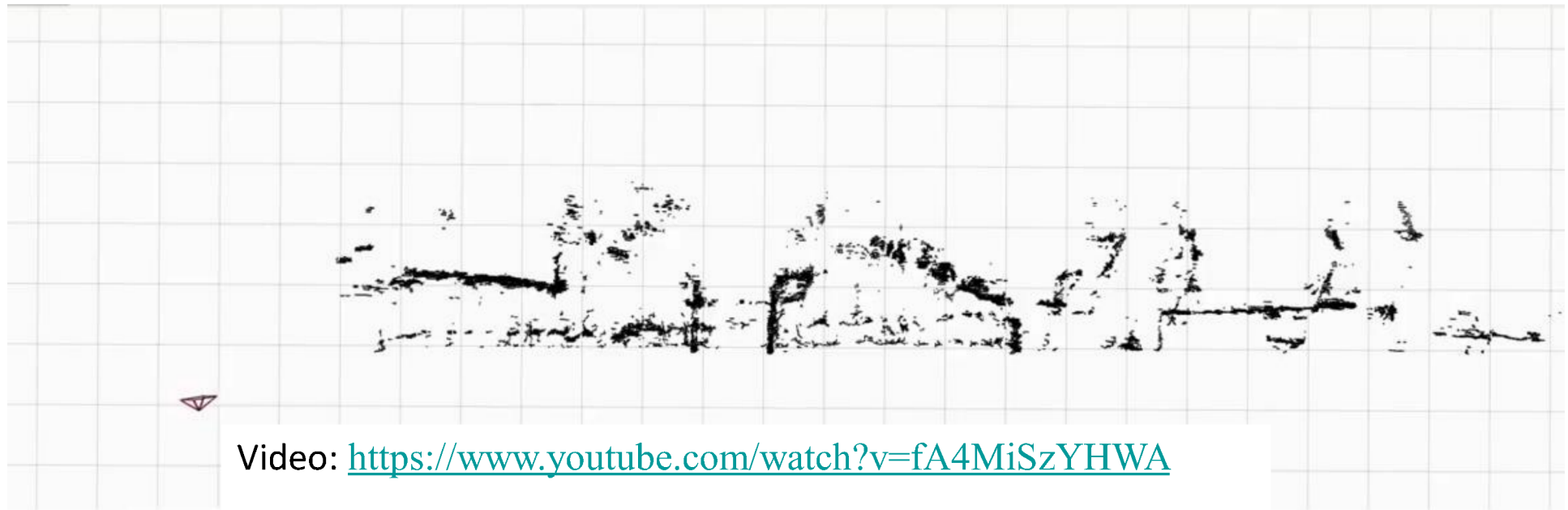
Application 1: Image Stabilization

- Problem: Estimate rotational motion (3DoF) of an event camera
- Can process millions of events per second in real time on a smartphone CPU (OdroidXU4)
- Works up to over **~1,000 deg/s**



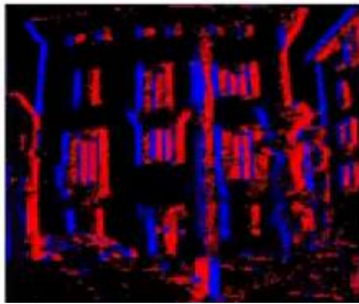
Gallego et al., Accurate Angular Velocity Estimation with an Event Camera, IEEE RAL'16. [PDF](#). [Video](#).

Application 3: 3D Reconstruction from a Train at 200km/h

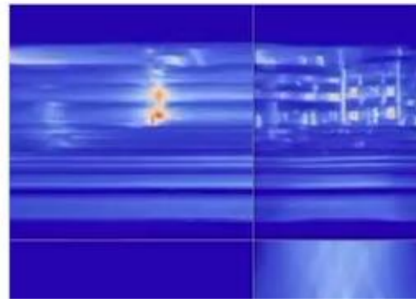


Video: <https://www.youtube.com/watch?v=fA4MiSzYHWA>

DVS
Events



DSI



External
camera



In collaboration with **SIEMENS** and the Swiss Railway company, SBB



Rebecq et al., EVO: A Geometric Approach to Event-based 6-DOF Parallel Tracking and Mapping, RAL'17. [PDF](#). [Video](#)

Rebecq et al., EMVS: Event-based Multi-View Stereo, IJCV, 2018. [PDF](#) [Video](#) [Source Code](#)

Application 4: Motion Segmentation

Conventional Frames



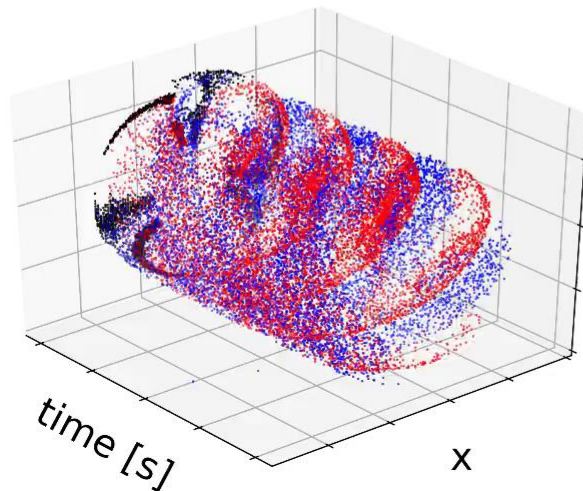
Sequence: Fan and Coin

One motion model is used per cluster; one for the fan, modelling rotation, one for the coin, modelling optic flow

Motion-Compensated Segmented Events



Events



Stoffregen et al., Motion Segmentation by Motion Compensation, arXiv 2019. [PDF](#). [Video](#).